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Advancement in Graph Neural Networks for EEG Signal Analysis and Application: A Review

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ABSTRACT Electroencephalography (EEG) can non-invasivey measure neuronal events and reflect brain activity at different locations on the scalp. Early studies for EEG signal processing have focused more on extracting EEG temporal features and considered the topology of EEG channels less due to the limitation of rich spatial information. Graph neural networks (GNNs), as a new kind of deep learning method, can use EEG signals as graph vertices, capturing the hidden topological connections between signals. GNNs have made great progress in EEG studies due to the advantage. In this overview, we review the very new and fundamental models of GNNs and their modifications, such as graph regularized neural networks, graph convolutional neural networks, spatial-temporal graph neural networks, graph attention networks, and their variants in EEG signal analysis fields. The applications of GNNs are summarized in the domains of emotion detection, epilepsy seizure detection, stroke rehabilitation, Alzheimer's disease diagnosis, motor imagery detection, neurological disease diagnosis, major depressive disorder, and driving fatigue detection. We employed a Systematic Literature Review (SLR) approach to select 79 papers for a comprehensive review. The current state is analyzed and forecasts are provided based on the available difficulties. We conclude by suggesting potential directions for future research in this rapidly developing topic.

INDEX TERMS Electroencephalogram, Deep Learning, Graph neural networks, Variants of Graph neural networks, Applications of Graph neural network.

I. INTRODUCTION

Electroencephalography (EEG) is a non-invasive technique that is used to measure and record the brain's electrical activity through the electrodes placed on the scalp. The brain's neurons communicate by generating small electrical impulses [1]. Its real-time monitoring capabilities make it a widely used tool in neuroscience, psychology, clinical diagnostics, and brain-computer interface (BCI) development [2]. There are several types of EEG technology, such as spontaneous EEG, event-related potentials (ERPs), induced EEG (elicited by stimuli but not time-locked), EEG-based brain-computer interfaces, and high-density EEG, among others. EEG has significant applications in detecting and diagnosing neurological conditions such as epilepsy, emotion, sleep disorders, neurological disease and brain injuries, as well as in research focused on understanding cognitive processes and emotional states. Despite its limitations in spatial resolution, EEG remains a non-invasive, affordable, and effective method for studying the brain, providing crucial insights into its functioning and supporting advancements in both clinical and research domains [3, 4].

Machine learning approaches to EEG analysis include traditional step-by-step methods and end-to-end deep learning [5]. Initially, techniques like Common Spatial Patterns (CSP) [6], wavelet transforms (WT) [7], and Hilbert-Huang transforms [8], were used to analyze the brain's regional signals and neural temporal dynamics. Power Spectral Density (PSD) [9], another crucial technique, helps define various brain states. However, the reliance on manually defined features limits traditional classifiers. The adoption of automated feature extraction is crucial for enhancing classifier adaptability, reducing bias, and improving efficiency across diverse EEG datasets.

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Deep learning (DL) architectures are becoming increasingly utilized in EEG signal analysis, enhancing feature integration and classification over traditional classifiers [10]. Despite their growing popularity, two significant limitations persist: high processing costs and the need for large, diverse datasets to fulfil their data-intensive nature. Strategies to mitigate these include reducing parameters, inputting raw EEG data, reusing features, and enhancing feature learning, with recent advancements addressing long training times [11]. Among the DL frameworks, long short-term memory (LSTM) networks [12], Deep belief networks (DBNs) [13], Deep neural networks (DNNs) [14], and convolution neural networks (CNNs) [15] have shown improvements in feature extraction and EEG data classification. CNNs, in particular, simplify network complexity and reduce the need for extensive feature extraction and data reconstruction [16]. However, they struggle with non-Euclidean data due to the limitations of discrete convolution in maintaining translational invariance, which impedes their ability to fully utilize neighbor information in graph structures.

Graph Neural Networks (GNNs) have gained consideration for their ability to enhance feature extraction, pattern identification, and classification across various domains, including EEG signal analysis. Unlike traditional methods, GNNs model data as graphs, where nodes represent EEG electrodes, and edges capture the relationships or interactions between them, which can be weighted or unweighted based on neuroscientific insights [17]. This approach enables GNNs to effectively capture the complex interrelationships between brain regions by processing brain connectivity as a graph structure. Research conducted by [18], illustrates the adaptability and variety of graph representations in various disciplines, including molecular chemistry, data mining, computer vision, and pattern recognition. This graph-based technique allows GNNs to detect subtle patterns and anomalies in brain activity that may be skipped by conventional learning methods in EEG applications.

EEG is a valuable tool for studying selective attention. For instance, top-down influences in the alpha (~8–12 Hz) and beta (~13–30 Hz) frequency bands are crucial for regulating selective attention [19]. EEG-based GNNs offer deeper insights into the dynamic interactions between brain regions, providing significant advantages in the study of selective attention [20]. Inspired by the attention mechanism, a novel attention-guided graph structure learning network based on EEG signals was proposed [21], improving the performance of auditory attention detection and holding promise for the development of neuro-steered hearing aids. As researchers highlight, the ability of GNNs to model brain connectivity makes them a transformative tool for EEG research, advancing progress in cognitive science, neurophysiology, and brain-computer interfaces [22].

Continued research in GNNs has led to the development of numerous variants, which are extensively applied in EEG signal classification, particularly within EEG-based applications. Several comprehensive review papers have been published that focus on the utilization of GNNs in this specific area of research, highlighting the significant contributions. Specifically, [23, 24], offers an extensive overview of GNNs tailored for EEG classification. In [25], this paper provides another comprehensive review of research focusing on neural networks for EEG signal processing. In [26], the authors introduce commonly used GNN-based algorithms and systematically review their applications in EEG-based disease diagnosis. They provide some basic models of GNNs and their variants, including graph convolutional neural networks, describe attentional networks, and summarize applications. Research in [27], proposes a general design process for GNN models, discussing variations and systematically categorizing applications across different fields. In [28], this review examines GNNs, a particular family of computational techniques, in interpreting EEG data. In [29], This article provides a comprehensive overview of GNNs in supervised, unsupervised, semi-supervised, and self-supervised learning. Notably, the previous reviews have not extensively addressed newly developed variants of GNNs and their applications in EEG-based settings.

The theme of the review paper focuses on the latest GNN variants and their applications in EEG signal analysis. It addresses the complexities of EEG data, which necessitate advanced analytical methods beyond traditional machine learning techniques. The paper shows that GNNs can handle the non-Euclidean structure of brain connectivity, which makes it much easier to find subtle neurological patterns and problems and fills in the gaps left by earlier research.

Key points of this review include:

- Evaluating cutting-edge and fundamental graph-based neural network models and their adaptations for EEG classification.
- Concentrating on newly developed GNN variants tailored for EEG data classification.
- Reviewing the wide spectrum of GNN applications in classifying all types of EEG signals.
- Providing comprehensive details on new variants, dataset specifics, publication data, and analytical comparisons of studies.

This review aims to serve as a valuable resource for researchers new to the field of GNNs by offering an extensive overview of state-of-the-art graph-based models and their applications in EEG studies. The paper discusses the methodology for literature selection, provides an overview of EEG signals, GNN architectures suitable for EEG, and the application scenarios of GNNs in EEG

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analysis, and evaluates current limitations while proposing future prospects for GNNs.
In Section 2, we discussed the search and paper selection process. Section 3 presents a brief discussion about EEG signals. In Section 4, we explored the GNN network structure and its variations for EEG analysis. Section 5 focuses on the application scenarios of graph neural networks in analyzing EEG data. Finally, in section 6, we examine the shortcomings of current approaches, discuss the prospects and future work for GNNs, and provide conclusions.

II. METHODOLOGY OF SURVEY

In this section, we will discuss the selection method of articles by using the Systematic Literature Review (SLR) approach [30]. The process involves collecting articles from several repositories, removing duplicates, applying snowballing techniques, and analyzing abstracts, introductions, and conclusions. Articles not meeting the inclusion criteria are excluded, while the remaining undergo full-text analysis and quality assessment. The entire process is shown in Fig. 2. This systematic approach ensures the selection of high-quality and relevant studies. Any discrepancies during the selection process were resolved through consensus among the authors.

A. SEARCH AND SELECTION OF LITERATURE

We employ various search techniques to identify suitable papers, following three steps. Firstly, we determine search strings based on predefined criteria. Next, we select literature repositories. Finally, we execute the search using the chosen search strings in the selected archives.

1) SEARCH STRING

In Table I, we highlighted some necessary keywords from the inclusion criteria. Then, we use the keywords and their abbreviations and the Boolean operator "AND" to generate search strings. Keywords and abbreviations are electroencephalogram, EEG Classification, Graph neural network, Graph convolution network, and Graph neural network application. We use seven strings to find the appropriate publications.

TABLE I
SEARCH STRINGS USED TO FIND LITERATURE

String ID	Search String
String 1	“Electroencephalogram or Electroencephalography” AND “Graph neural network” AND “Application”
String 2	“Electroencephalogram or Electroencephalography” AND “Graph convolution network” AND “Application”
String 3	“EEG” AND “Graph neural network” AND “Application”
String 4	“EEG,” AND “Graph neural network,” AND “Deep Learning”
String 5	“EEG” AND “Classification” AND “Graph neural network”
String 6	“EEG” AND “Classification” AND “Graph convolution network”
String 7	“EEG”, AND “GNN”, AND “Deep Learning”

2) REPOSITORIES AND SEARCH OF PRIMARY LITERATURE

Using the SLR approach, we identified relevant papers through a comprehensive search of several publicly available repositories, including PubMed, IEEE Xplore, Science Direct, Research Gate, and The National Science Foundation. For our literature review, seven predefined search strings were applied across these archives, focusing on works published between 2018 and December 30, 2024 (as shown in Fig. 1). The publicly available tools PubMed, IEEE Xplore, Science Direct such as a custom-built Python script: PubMed (https://github.com/tatsuya-takahashi/PubMed-API-Script), IEEE Xplore (https://github.com/QY7/ieee_xplore), Science Direct (https://github.com/ElsevierDev/elsapy). Following the initial search, duplicate entries were removed, and the inclusion/exclusion criteria were applied. To ensure comprehensive coverage, forward and backward snowballing techniques were used to identify related publications, broadening the scope of the literature review through archive searches and reference retrieval.

B. THE INCLUSION AND EXCLUSION STANDARDS

In this part, we first set some specific standards according to SLR. Then we define some inclusion (EEG signal should be used as input data, graph based neural network methods, the structure of the model should be declared, and published in high-ranking journals or conferences within five years) and exclusion (EEG based but don’t use Graph neural network or its variants, use GNN but not EEG based, writing in language not in English, Unpublished) standards to selecting papers for review.

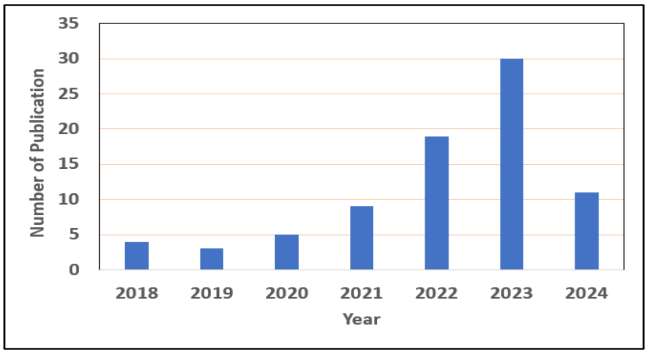

FIGURE 1. Depicts the distribution of papers over the years, highlighting a significant number of publications from 2021 and 2024 to emphasize the most recent advancements. Additionally, it includes the selected papers from the period of 2018 to 2020 to provide a comprehensive view of the field's evolution. This assessment was taken till December 30, 2024.

C. PRIMARY SELECTION AND QUALITY ASSESSMENT

After collecting the literature by using search strings and repositories, duplicates were removed. Then primarily the abstract, introduction, and conclusion of each paper were then reviewed to apply the inclusion and exclusion criteria. Publications meeting any exclusion criteria were immediately discarded. There are five inclusion criteria listed, and each of them was given a score of 0.5; if a publication had an overall

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score of 2 (meaning it met at least 4 of the five inclusion criteria), it was chosen for full-text analysis. This full text analyzed publication will further checked by quality and bias assessment criteria for the final selection.

D. BIASED QUALITY ASSESSMENT ANALYSIS

The analysis of the selected literature can sometimes be biased. While the inclusion and exclusion criteria may fulfil the objective of this review, then the quality assessment can have bias. The selected publications may not always be reliable, comprehensive, or have clear results and practical applications. These problems were mitigated by having separate inclusion and exclusion criteria and quality assessment criteria. As explained in section B (3) inclusion criteria were further filtered to assess quality. However, 79 articles that were finally chosen were thoroughly read and analyzed to confirm they met the quality assessment standards.

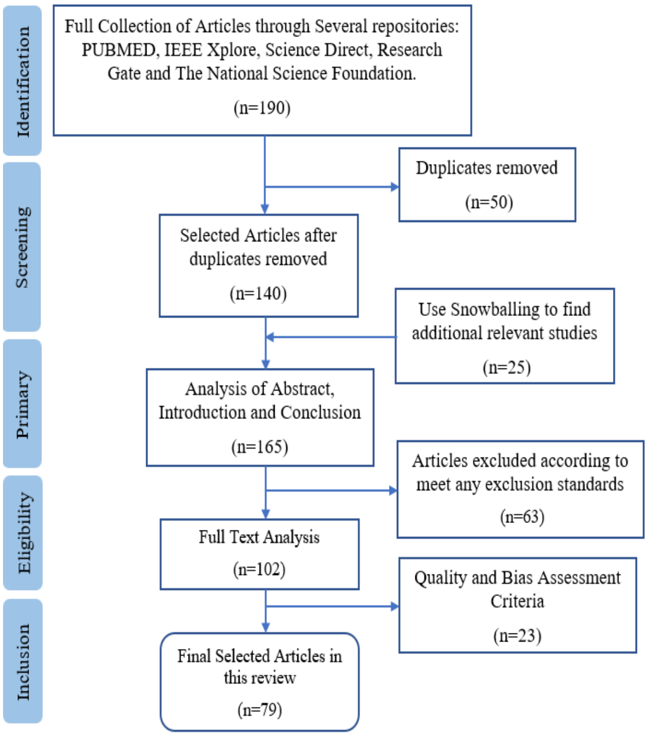

FIGURE 2. Article Selection Workflow for Systematic Review.

E. WORKFLOW FOR SYSTEMATIC REVIEW

The Fig. 2. illustrates the systematic process of article selection in this review, beginning with the identification phase, where 190 articles were collected from various repositories, including PubMed, IEEE Xplore, ScienceDirect, ResearchGate, and the National Science Foundation. After removing 50 duplicate articles, 140 articles were retained for further screening. An additional 25 relevant studies were identified through snowballing techniques, resulting in 165 articles for abstract, introduction, and conclusion analysis. During this phase, 63 articles were excluded based on exclusion criteria, leaving 102 articles for full-text analysis. These were subjected to quality and bias assessment, after which 23 articles were excluded, leading to a final selection of 79 articles for inclusion in this review. This systematic approach ensures a comprehensive and high-quality selection of literature for the study.

III. EEG SIGNAL ANALYSIS BY GNNs

EEG is a non-invasive neuroimaging technique used to measure and record electrical activity in the brain. This process involves placing electrodes on the scalp, which detect electrical signals generated by neurons as they communicate and transmit information. These signals are then amplified and visualized as wave patterns, known as EEG traces, which represent the brain's electrical activity. EEG captures various brainwave patterns classified into different frequency bands, including delta, theta, alpha, beta, and gamma waves, each associated with specific brain functions and states of consciousness. [31]. The frequency bands of EEG are associated with various cognitive and physiological states, though these associations are not indicative of direct causality. Delta waves (0.5-4 Hz) are often linked to deep sleep and certain neurological conditions. Theta waves (4-8 Hz) typically appear during drowsiness daydreaming, and in young children. Alpha waves (8-13 Hz) are commonly observed when an individual is awake but relaxed and not actively processing information. Beta waves (13-30 Hz) are usually present during wakefulness and active mental engagement, such as problem-solving and decision-making. Gamma waves (30+ Hz) are associated with higher-level cognitive functions, including memory and perception [32, 33]. EEG is widely used in clinical and research settings to diagnose and monitor conditions related to brain function, such as epilepsy, sleep disorders, and brain injuries, and to study cognitive processes in neuroscience research.

The EEG signals, with their unparalleled temporal resolution, are pivotal for deciphering brain activity and diagnosing neurological disorders. Yet, their intricate, non-Euclidean structure coupled with pervasive noise and overlapping frequency bands, poses formidable challenges for conventional decoding techniques. Through the joint efforts of numerous researchers, EEG signal analysis has evolved from applying method variants from other fields to becoming a fully original and challenging domain. Given that EEG signals are collected from the human scalp, with electrode distribution approximating a spherical surface, GNNs can be effectively utilized, securing an important position in this field. GNNs are a type of neural network that can model and analyze data represented in graph structures, making them suitable for EEG analysis due to the inherent graph-like nature of brain connectivity [34]. In Fig. 3, provides a general design pipeline for structure GNN models, which includes data preprocessing, graph representation construction, and model architecture design.

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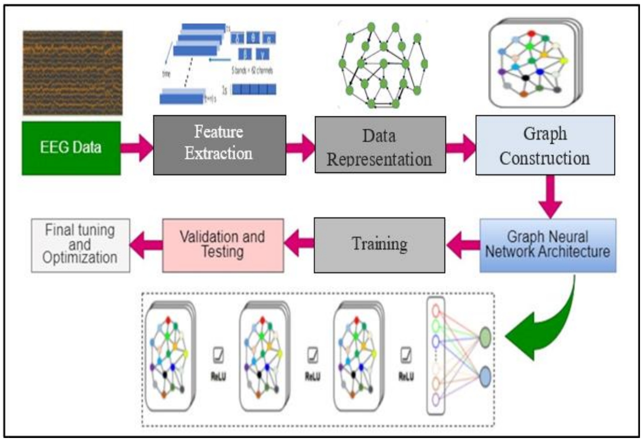

FIGURE 3. Architecture of a Graph Neural Network for EEG data analysis. The workflow begins with EEG Data, which undergoes Feature Extraction to generate relevant characteristics from the raw signals. The extracted features are then Data Represented, transforming them into a graph structure. Graph Construction creates a network of nodes (representing EEG channels or segments) and edges, based on relationships such as proximity or similarity. The GNN architecture incorporates layers like multi-head attention (ReLU activations) to focus on the most important node connections, resulting in enhanced predictions for EEG analysis. Afterward, the model is trained using the graph-based data, with Validation and Testing for evaluation. Finally, the network undergoes Final Tuning and Optimization to improve performance.

Here's a general approach to EEG analysis using GNNs:

1) Feature Extraction: Extract features from the EEG signals recorded at each electrode. These features include amplitude, frequency-based features, and statistical measures,etc.
2) Data Representation: EEG data can be represented as a graph, where nodes represent specific EEG channels, and edges represent the functional or structural connectivity between these channels.
3) Graph Construction: Construct a graph where electrodes are nodes, and the edges between nodes represent functional connectivity measures [35]. The edges can be based on correlation, coherence, or other connectivity metrics derived from EEG signals.
4) Graph Neural Network Architecture: Design a GNN architecture that takes the EEG graph as input and processes it to extract relevant features and learn the underlying patterns in the connectivity data. GNNs can learn node and edge representations, capture spatial and temporal patterns, and model the dynamics of brain activity [36].
5) Training: The Graph Neural Network: Train the GNN using labeled data, where the labels could represent different states or conditions.
6) Validation and Testing: Evaluate the trained GNN on unseen data to assess its performance in various EEG analysis tasks.
7) Optimization: Fine-tune the GNN architecture and hyperparameters to improve performance and efficiency

Applying GNNs to EEG data enables researchers to gain insights into brain connectivity, understand brain disorders, develop diagnostic tools, and potentially discover new treatments. It's a promising area that's still evolving, and ongoing research continues to improve the accuracy and interpretability of EEG analysis using GNNs.

IV. VARIANTS OF GRAPH NEURAL NETWORKS FOR EEG SIGNAL PROCESSING

In this section, we will discuss the basic graph neural network architecture used in EEG-based research, and its variants, which will help us to explain and understand the application state in the next segment.

A. GRAPH NEURAL NETWORK

GNNs are specialized neural networks designed to operate on graph data structures, predicting nodes, edges, and graph-based tasks. They are particularly effective for tasks like graph classification and utilizing EEG recordings to construct graphs and make predictions. GNNs extend traditional neural network operators to graphs, lacking a clear sense of distance or order like images, with graph convolutional layers being a key component. These layers employ a message-passing mechanism, updating each node's representation by aggregating messages from neighboring nodes [37]. This methodology is analogous to image convolution, allowing the features of nodes to disseminate across the graph effectively.

A graph is a topological space generated by the triple $G = (V; E; W)$ of vertices, edges, and weights, where V represents the set of nodes (i.e., EEG electrodes/channels), E represents the set of edges, and W is the weight that represents the adjacency relationship between vertices or nodes [38]. The adjacency matrix of an undirected graph must be symmetric, $W_{ij} = W_{ji}$. The feature vector of a node is defined as $X \in R^{n\times d}$, where X_{v_i} is a feature node of v_i. In general, after following all the previous steps, the following expression is obtained:

$$H_t = \sigma\left(D_{IN}^{\frac{-1}{2}} A^T D_{IN}^{\frac{-1}{2}} H_{t-1} W_t\right) + b_t \tag{1}$$

Where H = Matrix of node Feature, A is Adjacency Matrix, D is Degree Matrix, W t is trainable weights Matrix and σ denotes Activation function.

Recent studies have utilized GNNs for various EEG-based applications shown in Table II, demonstrating their versatility in signal classification and disease detection. In [39], researchers applied a GNN model to classify EEG signals, represented as channel samples collected during trials. Explored the use of GNNs for fatigue detection from EEG data. Efforts to improve interpretability in epilepsy detection were made by integrating graphic representations of EEG signals with GNN models [40, 41]. A hierarchical aggregation-based GNN was employed to analyze the minimum spanning tree structure in EEG-based emotion recognition [42]. Further developments include the proposal of a GNN-based framework for representing EEG data as graph signals, utilizing GNN Explainer to identify critical subgraph structures for classification decisions [40]. This tool was also applied in a study involving mental arithmetic tasks to discover pertinent relationships [43]. Additionally, a specialized GNN variant, the Self-Organized Graph Neural

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Network (SOGNN), was developed for cross-subject emotion recognition from EEG data [44].

1) REGULARIZE GRAPH NEURAL NETWORK (RGNN)

This is another extension idea on GNN, as shown in Table II. Some researchers have developed this extension in the area of EEG-based emotion recognition. The RGNN model is biologically supported and captures both local and global relationships between channels. To increase the model's resistance to cross-subject EEG fluctuations and noise labeling during recording, they proposed two regularizes, NodeDAT and EmotionDL [45]. The biologically-supported adjacency matrix and two regularizes improve the model's performance significantly. The adjacency matrix $A_{ij} \in R^{n \times n}$represents the topological organization of channels on the brain surface. Here, n denotes the number of channels, A_{ij} is the adjacency matrix, which denotes the weight of the connection between channels i and j . Here, $\mathrm{A_{ij}}$ Overs self-loop. They design $\mathrm{A_{ij}}$ as a symmetric due to avoiding overfitting by using $\frac{\mathrm{n(n+1)}}{2}$ parameters instead of n^2 . Functional connection between brain areas deteriorates as an inverse square or gravity law function of physical distance. So, the adjacency matrix is as follows,

$$A_{ij} = exp\left(-\frac{d_{ij}}{2\delta^2}\right) \tag{2}$$

Where, d_{ij} Denotes the physical distance between channels, and δdenotes a sparsity hyper-parameter. address the, Node-wise Domain Adversarial Training (NodeDAT) enhances subject-independent classification by reducing disparities between source and target domains. A domain classifier categorizes node representations as source (0) or target (1), and the model learns domain-invariant representations by minimizing binary cross-entropy losses.

$$\emptyset_{\boldsymbol{D}} = -\sum_{i=1}^{N}\sum_{j=1}^{n}\left(\log\left(p_j(0|X_i^S,\theta_D)\right) + \log\left(p_j(1|X_i^T,\theta_D)\right)\right) \tag{3}$$

Where, 0 and 1 denote source and target domains, respectively, θ_D denotes the parameters of the domain classifier. To handle noisy emotion labels, Emotion-Aware Distribution Learning (EmotionDL) learns class distributions instead of single labels. The model optimizes by minimizing Kullback-Leibler (KL) divergence.

$$\emptyset' = \sum_{i=1}^{n} KL\big(p(Y|X_i,\theta),\hat{Y}_i\big) + \alpha\big||A|\big|_1 \tag{4}$$

Where, $p(Y|X_i,\theta)$ denotes the output probability distribution. By combining NodeDAT and EmotionDL, the overall loss function of RGNN can be rewritten as,

$$\emptyset'' = \emptyset' + \emptyset_{\boldsymbol{D}} \tag{5}$$

2) GRAPH GENERATIVE NEURAL NETWORK (GGN)

Currently, GNNs have become increasingly popular in EEG analysis, with ongoing research focusing on improving and developing new variants. A recent sub-variant, open GGN, was introduced in [46], to dynamically discover brain functional connections from EEG data of epilepsy patients exposed in Table II. The GGN model features two main modules: a connectivity graph generator and a spatial decoder. The connectivity graph generator consists of Para-Learner, Mix-Gauss module, and Gumbel-Sampler components to model changing connectivity over time. Meanwhile, the spatial decoder extracts spatial information to create node-level representations for classification. This innovative approach shows promise in epilepsy seizure recognition and classification tasks. If the data set collected from the brain surface is represented by $(X, Y, A^{(0)})$, where X is a time series sample, Y denotes the corresponding labels, and $A^{(0)}$ is the initial connection weight. Let's be the sample index in the training data and θ is the collective set of all learnable parameters. The likelihood function for the GGN architecture optimization objective can be rewritten as:

$$L\big(\theta;\{X,Y,A^{(0)}\}\big) = \prod_{s=1}^{N}\sum_{m=0}^{M} P_\theta\big(Y_s A^{(m)}|X_s A^{(0)}\big) \tag{6}$$

B. GRAPH SEQUENCE NEURAL NETWORK (GSNN)

Graph neural networks are now a common tool in EEG analysis. GSNN is a novel kind of GNN [47] that proposed a GSNN that accurately extracts motor image patterns from EEG in the presence of distractors. GSNN aims to construct subgraphs exploiting the biological topologies of brain regions to capture local and global relationships between characteristic channels. The GSNN concept adopts the graph convolutional network with a self-monitoring mechanism and a nodal domain adversarial. A node-level adversarial training method to improve the classification accuracy of each proposed task Self-adaptability of GSNN from a brain channel perspective.

The model's goal during optimization is to confuse the domain classifier by learning domain-invariant representations for each node. The domain classifier seeks to minimize the two binary cross-entropy losses shown below.

$$\emptyset_D = -\sum_{i=1}^{B}\sum_{j=1}^{N}\left(\log\left(p_j\big(0|X_i^0,\theta_D\big)\right) + \log\left(p_j\big(1|X_i^E,\theta_D\big)\right)\right) \tag{7}$$

Where 0 and 1 denote source and target domains, respectively, θ_D denotes the parameters of the domain classifier, E is an unlabeled target p_j is the domain probability. Once the converted class distributions are obtained, the model may be optimized by reducing the following Kullback-Leibler (KL) divergence.

$$\emptyset' = \sum_{i=1}^{n} \mathrm{KL}\big(\mathrm{p}(\mathrm{Y}|\mathrm{X_i},\theta),\hat{\mathrm{Y}}_i\big) + \alpha\big||\mathrm{A}|\big|_1 \tag{8}$$

Where, $\mathrm{p}(\mathrm{Y}|\mathrm{X_i},\theta)$ denotes the output probability distribution. By combining NodeDAT and graph sequence network, the overall loss function of RGNN can be rewritten as,

$$\emptyset'' = \emptyset' + \emptyset_{\boldsymbol{D}} \tag{9}$$

TABLE II
A BRIEF SUMMARY OF FINDINGS FROM SELECTED PUBLICATIONS OF GNN, GGNN, GSNN, GAT

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Reference	Year	Data	Model	Application	Classification Accuracy (Mean ± Std)
[38]	2022	Physionet: ErrP: RSVP	EEG-GAT	EEG classification	Accuracy: 58.09%, Accuracy: 76.42± 0.29%, Accuracy: 96.27±0.18%
[39]	2021	ErrP RSVP	EEG-GNN	EEG classification	Accuracy: 76.73±0.40% Accuracy: 93.49±0.10%
[40]	2023	CHB-MIT SSW	GNN	Epileptic Detection	Accuracy: 93.6% Accuracy: 95.8%
[41]	2023	CHB-MIT	GNN	Epileptic Seizure Classification	Average Accuracy: 87.54%
[42]	2022	DEAP	MST-GNN	Emotion classification	Accuracy: 92.31%/5.34%
[48]	2022	Home made	GNN	Alzheimer's disease	Accuracy: 92%, AUC: 98.84%
[49]	2023	SEED	GNN	Emotion Recognition	Accuracy: 98.35%
[50]	2021	Home made	GNN	Seizure Analysis	AUROC: 87.5%. F1 Score: 74.9%
[51]	2022	Homemade	GNNexplainer	Schizophrenia patients	Accuracy: 61.00 ±0.015%
[44]	2021	SEED SEED-IV	SOGNN	Emotion Recognition	Average Accuracy: 86.81±5.79% Average Accuracy: 75.27±8.19%
[45]	2019	SEED SEED-IV	RGNN	Emotion Recognition	Subject-dependent Accuracy: 94.24±05, 95% 79.37±10.54% Subject-independent Accuracy: 85.30±06.72%, 73.84±08.02%
[46]	2022	Home made	GGN	Epileptic seizure Detection	Accuracy: 91%
[47]	2022	Self	GSNN	Motor imagery	Recall: 80.44%, Precision=81.07%, F1 Score=80.5%
[52]	2023	Homemade	AGNN	Classification of autism	Accuracy: 97.1%, Precision: 95.1%, Specificity: 93.4%.
[53]	2023	DEAP SEED-IV	ResGAT	Emotion Recognition	Arousal: 87.06%, Valence: 89.26%, SEED-IV: 97.73%
[54]	2020	TUH	GAT	EEG Classification	Accuracy: 81.27%
[55]	2023	SEED SEED-IV DREAMER	STGATE	Emotion Recognition	Accuracy: 90.37% Accuracy: 76.43% Accuracy: 76.35%
[56]	2023	PD Oddball Data	ASGCNN	Parkinson's disease	Recall: 90.36%, Precision: 88.43%, F1-score: 88.41%, Accuracy: 87.67%, Kappa measures: 75.24%
[57]	2023	Homemade	GNN	Fatigue detection	Average Recognition: 87.5%
[58]	2023	Self-built	GNN	Fatigue driving recognition	Accuracy: 98%
[43]	2023	Open-access	GNNexplainer	Mental Arithmetic	Connectivity dataset: 85.57 ± 6.27% PSD and the connectivity: 96.26 ± 4.14%
[59]	2021	DEAP	GFT-STANN	Emotion Recognition	Valence: 94.8%, Arousal: 96.1%, Four Class: 92.7%
[60]	2023	BRC	CAGNN	Fatigue detection	Accuracy: 72.6%
[61]	2023	DEAP	SS4-STANN	Emotion Recognition	Valence: 95.6%, Arousal: 97%
[62]	2024	Clinical data	GNN GNN	Alzheimer's disease Seizures	Alzheimer's disease: Binary Accuracy: 91.77%, Ternary Accuracy: 89% Seizures: Binary Accuracy: 99.84%, Ternary Accuracy: 96.95%
[63]	2024	Clinical data	D-GNN Attention	Epilepsy detection	Average Accuracy: 99.83%, Specificity: 99.91%, Sensitivity: 99.78%, Precision: 99.87%, F1 Score: 99.47%
[64]	2024	SEED	Dynamic GNN	Emotion recognition	Average Accuracy: 94.35±5.05%
[65]	2024	DEAP SEED	GERBN	Emotion Recognition	Valence: 80.43%, Arousal: 88.47%, SEED: 92.37%

Table II offers a thorough overview of selected research papers that utilize graph-based algorithms and their various adaptations. It details the research methodologies employed, the datasets used, publication specifics, and the outcomes achieved in these studies. By encompassing information on research methods and dataset specifics, it highlights the diverse applications and advancements in these graph-based neural network algorithms. The table also provides publication details such as titles, authors, institution, and years, which helps readers locate the original sources. Additionally, the results section showcases the performance metrics and comparative analyses against baseline models, illustrating the effectiveness of the proposed methods. Overall, this table serves as a valuable resource for researchers and practitioners interested in the latest developments and applications of graph-based neural networks in various domains.

C. ADAPTIVE GRAPH NEURAL NETWORK (AGNN)

This is another new idea for graph neural networks, as shown in Table 2. In [52], in order to investigate abnormalities in the left inferior frontal gyrus among people who have Autism

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Spectrum Disorder (ASD) and typical development (TD), the adaptive graph neural network (AGNN) was proposed for classification purposes. They use it to mine the correlation of internal channels in the brain and better capture the spatio-temporal properties between channels. Here's the brain-modeled graph:

$$G = (V \times E \times A) \tag{10}$$

Where, $A \in R^{N \times N}$ is the adjacency matrix, V is the vertex set, E is the edge set, and $|V| = N$ represents the number of vertices. The graph space learning layer generates the adaptive graph adjacency matrix, which is then sent to the graph convolution module.

D. GRAPH ATTENTION NETWORK (GAT)

The introduction of GATs marked a pivotal point in this area, as shown in Table II. By employing attention mechanisms, GATs allow the model to focus on the most relevant parts of the input graph, improving the quality of the learned node representations. GATs adopt attention mechanisms to dynamically weigh the importance of neighboring nodes, enhancing the model's ability to focus on critical features [66, 67]. A graph attention network's input is a sequence of node feature vectors, which represented by H,

The attention weight among nodes can be calculated as,

$$a_{ij} = Softmax_j(e_{ij}) = \frac{exp\left(LeakyReLU\left(\vec{a}^T[W\vec{h}_i||W\vec{h}_j]\right)\right)}{\sum_{k \in N_i} exp\left(LeakyReLU\left(\vec{a}^T[W\vec{h}_i||W\vec{h}_k]\right)\right)} \tag{11}$$

Where, N_i denotes the neighborhood of node i, W is the weight matrix, $\vec{a}^T$ represents the transposition of the attention weight vector, and LeakyReLU denotes the nonlinear activation function. The graph attention network uses a multi-head mechanism to capture different information from each head, and the attention coefficients are combined with the corresponding feature vectors to compute the final output features of each node.

$$\vec{h}_i' = \sigma\left(\frac{1}{K}\sum_{k=1}^{K}\sum_{j \in N_i} a_{ij}^k W^k \vec{h}_j\right) \tag{12}$$

Where σ denotes the nonlinear activation function, $\vec{h}_i'$ is the final output vector of the graph neural network. A residual graph attention neural network, designed for multi-channel EEG emotion recognition, creates a sparse feature matrix from electrode channel positions, demonstrating the utility of attention-based GNNs in EEG research [53], The paper introduces STGATE, a spatial-temporal graph attention network incorporating a transformer-encoder. This network enhances emotion recognition by capturing time-frequency data and applying it to a spatial-temporal graph attention mechanism for more accurate classification.

In [56], attention-based graph neural networks are used to diagnose Parkinson's disease using EEG signals. The approach features a sparse graph convolutional neural network (ASGCNN), which integrates a graph structure to analyze channel relationships and uses an L1 norm for channel sparsity, enhancing the detection capabilities through attention mechanisms and spatial feature modeling.

In [63], authors propose a multilayer dynamic graph neural network with an attention mechanism for epilepsy detection.

E. GRAPH CONVOLUTION NEURAL NETWORK

Graph Convolutional Neural Networks (GCNNs) or Graph Convolutional Networks (GCNs) represent a neural network architecture designed specifically for handling graph-based data structures, where nodes denote entities and edges signify relationships between them. Widely applied in EEG-based research, GCNNs merge graph neural networks with convolutional neural networks to achieve notable success rates, as shown in Tables III and IV. By applying convolutional processes directly to graphs, GCNNs excel at extracting crucial features and patterns from interconnected nodes, proving invaluable in various applications where data naturally adopts graph structures, including social networks, biological networks, and data graphs [68].

By combining spectral graph theory with deep learning, GCNN enhance non-Euclidean data analysis for accurate classification. In GCNN, researchers explore convolution operations on graphs, presenting two strategies: spatial domain convolutions [69] and spectral domain convolutions [70]. Spectral-based methods use graph signal theory, employing transforms like Fourier or wavelet to shift graph signals to the spectral do-main, which is useful for filtering noise from EEG signals [71].

TABLE III
ALL GCNN VARIANTS AND EXPLANATIONS

Reference	Graph Convolution Neural Network Variants	Explanation
[72]	Adaptive Gated Graph Convolutional Network (AGGCN)	AGGCN, a variant of GCNN proposed for Alzheimer's disease diagnosis, integrates convolutional node feature enhancement with graph structures.
[73]	Coherence Based Graph Convolutional Network (C-GCN)	C-GCN analyzes motor imagery EEG data post-spinal cord injury, extracting temporal, frequency, and spatial properties alongside functional connectivity.
[74]	Fusion Graph Convolutional Network (FGCN)	FGCN extracts diverse relationships from EEG data by deriving brain connection features from topology, causality, and function, using a local fusion technique to utilize channels effectively.
[75]	Linear Graph Convolution Neural Network (LGCN)	LGCN preprocesses raw EEG signals and combines resulting matrices into time-frequency blocks for feature selection. It comprises two layers where each node aggregates attributes from first-level nodes in the first layer.
[76]	Multi-granularity Graph Convolution Network (MGGCN)	MGGCN utilizes multiple threshold values to construct a multi-grain functional neural network, preserving useful weak connections while removing noise.

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[77]	Double-branch Graph Convolutional Attention Neural Network (DGCAN)	DGCAN uses a graph neural network to filter out channels less influenced by spatial location variables and then employs spatial-temporal domain convolution to extract features.
[78]	Multi-head residual Graph Convolutional Neural Network (MRGCN)	MRGCN decomposes multi-band differential entropy functions to capture temporal complexity in emotion-related brain activity and combines short-range and long-range brain networks to explore intricate topological characteristics.
[79]	Multi-scale Attentional Temporal Convolutional Neural Network (MATCN-GT)	The MATCN block directly derives features from EEG signals without prior knowledge, while the GT block focuses on EEG signal interactions among specific electrodes.
[80]	Progressive graph convolution network (PGCN)	PGCN utilizes a dual graphics module to leverage interaction between multiple EEG channels, combining dynamic functional connectivity and static geographical proximity data of brain areas to match distinct EEG patterns.
[81]	Spatial-temporal graph convolutional neural network (ST-GCN)	ST-GCN fully uses functional connectivity's constrained spatial topology as well as the discriminative dynamic temporal information provided by 1D convolution.
[82]	Graph-embedded convolutional neural network (GECNN):	GECNN enhances emotion detection by integrating local and global functional data using attention mechanisms and dynamic graph filtering.
[83]	Granger Causality-Graph convolution neural network (GC-GCNN)	GC-GCNN method generates a causality matrix using Granger causality analysis and transforms it into an adjacency matrix adaptively, aligning it with human brain cognitive patterns
[84]	Segmentation adaptive spatial-temporal graph convolutional network (SAST-GCN)	SAST-GCN for P3-based video object identification. It segments P3 signal data based on video-induced processing phases, constructing brain network connections accordingly.
[75]	Linear graph convolution network (LGCN)	LGCN and DenseNet are proposed for seizure detection. Graph nodes represent raw EEG data, reflecting its feature vector, while an adjacency matrix is constructed using Pearson correlation analysis.
[85]	Epilepsy EEG Graph Convolutional Network (EGCN)	EGCN optimizes channel correlations for enhanced data extraction in epilepsy EEG analysis. It features a 5-layer graph convolutional network designed for classifying healthy and epileptic signals.
[86]	Sequential Graph Convolutional Network (SGCN)	SGCN uses Frequency-domain Complex Networks of EEG Signals for epilepsy detection, combining advanced GNN architecture with sequential convolution.
[87]	Dynamical Graph Convolutional Neural Networks (DGCNN)	DGCNN dynamically learns connections between multiple EEG channels through the adjacency matrix, allowing for more discriminative feature extraction.

Table III provides detailed explanations of various articles utilizing GCNN and their variants. It highlights the unique methodologies and applications of different GCNN models. This compilation is a valuable reference for understanding the diverse implementations and effectiveness of GCNN variants in solving complex problems across various domains.

1) SPECTRAL-DOMAIN GRAPH CONVOLUTION NEURAL NETWORK

Spectral-based GCN is a successful CNN transformation that employs the graph signal approach with the Fourier transform, or alternatively, a wavelet transforms as a bridge, and graph signals are transformed from the vertex domain to the spectral domain. This may be accomplished using either the graph Laplacian or the normalized Laplacian matrix to handle non-Euclidean structure data. This section first introduces spectral theory and describes how graph signal theory is used to transfer classical CNNs to graphs, followed by a typical GCN model in the spectral domain [88], establishes a theoretical foundation for convolution in the spectral domain, with practical applications and initial approximations presented in [89]. From the mathematical equation spectral graph convolution is defined as follows,

$$X_{k+1,j} = h\left(U \sum_{i=1}^{f_{k-1}} F_{k,i,j}\, U^T X_{k,i}\right)(j = 1 \dots f_k) \qquad (13)$$

Where, $X_{k+1,j}$ denotes the convolution layer and $X_{k,i}$ represents node feature, f_{k-1} denotes the dimension of the output feature, h is the activation function, U is the vector that contains the eigenvectors for each eigenvalue matrix. GCNNs are increasingly utilized in EEG research, especially for disease detection such as epilepsy. In [85], a specialized 5-layer GCNN architecture was developed to classify EEG data from healthy and epileptic patients, showcasing its effectiveness in medical diagnostic.

TABLE IV
A BRIEF SUMMARY OF FINDINGS FROM SELECTED PUBLICATIONS OF GCNN

Reference	Year	Data	Model	Application	Classification Accuracy (Mean ± Std)
[90]	2022	SEED DEAP	DGCNN	Emotion Recognition	SEED: (Subject-dependent: 99.25%, Subject-independent: 98.51%) DEAP: (Subject-dependent: 98.96% Subject-independent: 98.31%)

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[91]	2022	Home made	GICN	Depression Recognition	Accuracy: 96.50%
[85]	2018	Bonn	EGCN	Epilepsy classification	Accuracy: 98.35%
[92]	2019	DEAP SEED	P-GCNN	Emotion Recognition	Valence: 73.31%, Arousal: 77.03%, Dominance: 79.2% SEED: 84.35%
[86]	2020	Bonn SSW	SGCN	Epilepsy classification	Average Accuracy: 89.3%
[93]	2022	SEED SEED-IV	CGCNN	Emotion Recognition	Average accuracy: 93.36% Average accuracy: 75.48%
[94]	2023	Neonatal EEG	GCNN	Seizure's detection	AUC90 of 96%, 95.7%, and 94.9%.
[72]	2023	Home made	AGGCN	Alzheimer's Disease	Accuracy: 89.61 ± 1.54%
[73]	2023	Home made	C-GCN	Spinal cord injury (SCI)	Accuracy: 96.85%.
[74]	2023	SEED SEED-IV	FGCN	Emotion recognition	Accuracy: 94.1% Accuracy: 77.14/15.71%
[75]	2023	CHB-MIT	LGCN	Epileptic Seizure Detection	Accuracy: 98%, Specificity: 98.60%
[76]	2023	MODMA MPHCE EDR	MGGCN	Major depressive disorder (MDD)	Accuracy: 99.68 ±0.43% Accuracy: 98.44 ± 0.78% Accuracy: 96.72 ±1.50%
[77]	2023	BCI IV 2a ,2b	DGCAN	Motor imagery	Accuracy: 84% Accuracy: 86%.
[78]	2023	DEAP SEED	MRGCN	Emotion recognition	Arousal: 95.72%, Valence: 94.97%, SEED: 98.9%
[95]	2023	Public data	GNN	Stroke rehabilitation BCI	Accuracy: 88.89%
[96]	2023	MASS-SS3 ISRUC-S3 SEED	BayesEEGNet	Sleep stage classification and emotion recognition	Accuracy: 85.56% Accuracy: 79.88% Accuracy: 82.50%
[97]	2023	Hospital data	GCNN	Schizophrenia recognition	Accuracy: 90.01%.
[98]	2023	DEAP-Twenty DEAP-Geneva SEED	GCNN	Emotion recognition	Average Accuracy: 90.74%, Average Accuracy: 91%, Average Accuracy: 90.22%
[99]	2023	GigaScience	GCNN	Motor imagery detection	Accuracy: 93.73%
[100]	2023	SEED DEAP	STFCGAT	Emotion recognition	SEED: (Subject-dependent: 99.11%± 0.83%, Subject-independent: 94.83%±3.41%) DEAP: (Arousal: 91.19% ± 1.24%, Valence: 92.03% ± 4.57%)
[79]	2023	SEED-VIG	MATCN-GT	Fatigue driving detection	Accuracy: 93.67%
[101]	2023	PhysioNet High Gamma	GCNs-Net	Motor imagery (MI) tasks	PhysioNet: (Subject: 93.06%, Group Level: 88.57%), High Gamma: (Subject: 96.24%, Group Level: 80.89%)
[102]	2023	CHB-MIT-EEG SEE-EEG TUH-EEG	GCN-LSTM GCN-BRF	Neurological Diseases Detection	CHB-MIT-EEG: (99.73%, 99.61%) SEE-EEG: (99.52%, 99.85%) TUH-EEG: (99.01%, 99.40%)
[80]	2023	SEED-IV SEED-V MPED	PGCN	Emotion recognition	Accuracy: 76.96±07.9% Accuracy: 71.40±09.43% Accuracy: 28.39±05.02%
[81]	2022	Home made	ST-GCN	Alzheimer's disease	Accuracy: 92.3%
[103]	2022	SEED DEAP	MdGCNN MdGCNN-TL-SC	Emotion Recognition	SEED: 91:54% and 85:01% DEAP: 86:67% and 70.19%
[104]	2022	SEED SEED IV DEAP	DBGC-ATFFNet-AFTL	Emotion Recognition	SEED: 97.31±1.47% SEED IV: 89.97±2.85% Valence: 95.91±3.27%, Arousal: 94.61±2.89%
[105]	2022	CHB-MIT	GCNN	Seizure Prediction	Accuracy: 96.51%
[106]	2022	Home made	GCNN	ESES	Accuracy: 91.2%, F1-Score: 95.0%, AUC: 96.5%, Sensitivity: 91.3%
[82]	2021	SEED SDEA DREAMER MPED	GECNN	Emotion Recognition	Average Accuracy: 92.93% Average Accuracy: 79.69% Average Accuracy: 95.73% Average Accuracy: 40.98%
[83]	2022	DEAP	GC-GCNN	Emotion Recognition	Arousal: 90.11%, Valence: 89.48%, Arousal-Valence: =86.35%
[107]	2022	SEED SEED-IV	MDGCN-SRCNN	Emotion Recognition	Average Accuracy: 95.08% Average accuracy: 85.52%
[108]	2022	DEAP	Mul-AT-RGCN	Emotion Recognition	Valence: 93.19%, Arousal: 91.82%
[84]	2022	Home made	SAST-GCN	Video target detection	Accuracy: 90.55%

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[109]	2021	Bonn SSW	SSGCNet	Epileptic EEG signal	Accuracy: 91.19%
[110]	2021	DEAP	GCNN	Emotion Recognition	Valence: 90.45%, Arousal: 90.60%
[111]	2021	CHB-MIT	LGCN	Seizure detection	Accuracy: 99.30%, Specificity: 98.82%, Sensitivity: 99.43%, F1 Score: 98.73%, AUC 98.57%
[112]	2020	TUH EEG MPI LEMO	EEG-GCNN	Neurological Disease Diagnosis	Average Accuracy: 85%
[113]	2021	MASS-SS3	ST-GCN	Sleep stage classification	Average Accuracy: 88.9%
[114]	2020	CHB-MIT TUH EEG	HGCN-TC	Epileptic Detection	5.77% Improve and 2.43% improvement
[87]	2018	SEED DREAMER	DGCNN	Emotion Recognition	SEED: (Subject-dependent: 90.4%, Subject-independent: 79.95% DREAMER: (Valence: 86.23%, Arousal: 84.54%, Dominance: 85%)
[115]	2019	Home Made	GCNN	Seizure classification	Better then RNN, CNN, or CNN
[116]	2022	ISRUC S3 ISRUC S1	JK-STGCN	Sleep Stage Classification	Average Accuracy: 83.1% Average Accuracy: 82.0%
[117]	2024	CHB-MIT	GCN-LSTM	Epilepsy Seizure Prediction	Binary: 99.39% Ternary: 98.69%
[118]	2024	GHB-MIT	GCN and Transformer	Epilepsy detection	Sensitivity: 92.97%, Specificity 94.60%, and Accuracy: 94.59%
[119]	2024	DEAP SEED SEED-IV	DAMGCN	Emotion Recognition	DEAP: (Valence: 96.96%, Arousal: 97.17%, Dominance: 97.50%), SEED: 99.42%, SEED-IV: 96.86%.
[120]	2024	PhysioNet	F-FGCN	Motor imagery	Subject: 96.11%, Group levels: 82.37%
[121]	2024	Home made	GCN	Fatigue driving detection	Case 1: 99%, Case 2: 97%, Case 3:96%, Case 4: 91%,
[122]	2024	SEED	SOGPCN	Emotion recognition	Subject-dependent = 95.26%, Subject-independent = 94.22%

Table IV details numerous papers utilizing GCNN and its variants, outlining the research methods, dataset specifics, publication information, and outcomes. This compilation serves as a valuable resource for new researchers, offering insights into various approaches and advancements in graph convolutional neural networks.

2) SPATIAL DOMAIN GRAPH CONVOLUTION NEURAL NETWORK

Spatial GCNNs have become favored in EEG classification, addressing the challenges posed by spectral-based GCNs in managing large datasets and processing graph structures directly. This spatial approach efficiently processes large-scale graphs by performing local graph convolutions on each node, which distributes complexity effectively [123]. It also facilitates the integration of diverse inputs through a merge function, enhancing its capability to handle complex data structures.

The concept of spatial GCNs was introduced [124], focusing on summarizing neighborhood information for graph convolution. This method involves generating a graphical representation by averaging node properties at each layer and updating it with the cumulative properties of all layers. By the design for the k^{th} layer, the output can be written as,

$$H^{(k)} = f\left(XW^{(k)} + \sum_{i=1}^{k-1} AH^{(k-1)}\theta^{(k)}\right) \tag{14}$$

The reading layer generates a graphical representation of each layer by averaging the properties of all nodes in each layer and updating the overall graphical representation with the sum property of each layer. Another advancement in spatial GCNs is described in [125], where EEG signals are segmented into frequency bands to construct intra-band graphs, which are then connected with inter-band links, improving the overall graph representation and addressing issues related to EEG's low dimensionality and spatial resolution.

V. APPLICATION OF GRAPH NEURAL NETWORK

GNNs analyze graph structures, which are ideal for real-world scenarios [112]. GNN are increasingly utilized in EEG systems, particularly in applications such as emotion detection, epilepsy etc. The distribution of publications across various EEG-based applications shown in Fig. 4. Ongoing research aims to enhance GNNs and develop new variants tailored for EEG signal analysis, addressing diverse neurological conditions for improved diagnosis and forecasting [126].

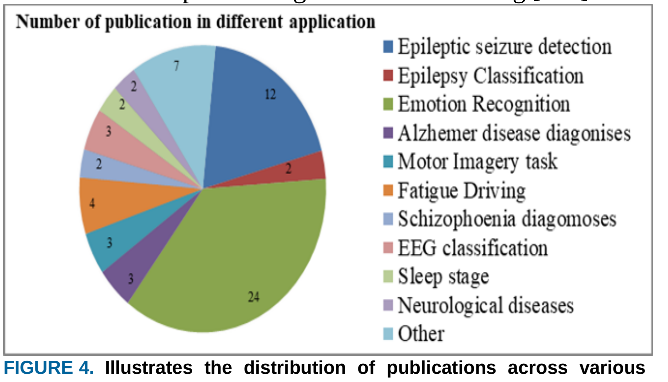

FIGURE 4. Illustrates the distribution of publications across various EEG-based applications. Emotion recognition-based studies comprised the majority, with up to 24 papers, while others addressed diverse EEG-related applications.

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A. EPILEPSY DETECTION

Epilepsy, one of the most prevalent neurological illnesses globally, is a chronic condition caused by episodic aberrant hypersynchronous electrical activity of neurons in the brain. Various brain disorders change the standard EEG picture, allowing the nature of the disease to be identified. Epileptic Seizure detection is popular research in the field of neurological diagnosis. Many researchers are using GNN to improve detection accuracy due to its high classification capability. Studies such as [40, 41, 46, 50] have applied GNNs to improve seizure detection, with [46], developing a weighted neighbor graph representation for EEG signals that enhances time and space efficiency. Further, GCNNs have been employed in several studies [94, 105, 109, 114], to successfully detect epileptic seizures, including the development of an automated neonatal seizure detection algorithm. Variant of GCNN, has also been used in [75, 111, 127], to preprocess EEG signals and facilitate feature selection for improved seizure detection. Additionally, novel methods have been explored, such as in [102], where a unique approach was developed for diagnosing seizures and other neurological diseases using EEG signals. Authors [86], they proposed Sequential Graph Convolutional Network that introduced SGCN to represent EEG signals in the frequency domain, enhancing data alignment and preserving sequential information. Another innovation, the Epilepsy EEG Graph Convolutional Network (EGCN) presented in [85], leverages channel correlations to extract significant data from EEG signals, demonstrating the robust potential of GCNNs in epilepsy detection. In [118], the authors proposed a hybrid model combining GCN and Transformer networks specifically for epilepsy detection from EEG signals.

B. EMOTION RECOGNITION

Emotion recognition involves identifying a person's emotional state using behavioral or physiological signals. In the realm of EEG-based emotion recognition, GNNs have gained prominence. Recent studies have focused on developing graph-based models for this purpose. Summarize various papers on graph-based networks for EEG-based emotion recognition, as shown in Table V. In a technique was introduced that encodes the connections between EEG signal sources into a GNN adjacency matrix, using a GNN classifier to detect emotional markers. Another study, [45], presents a RGNN that utilizes the biological topology of brain regions to analyze both local and global interactions between EEG channels. This involves leveraging the adjacency matrix of the graphical neural network, which is designed to reflect the connectedness and sparsity informed by neuroscientific principles of human brain architecture.

C. ALZHEIMERS DISEASE DIAGNOSES

As a neurodegenerative disorder, Alzheimer's disease (AD) is the most prevalent form of dementia. Patients with Alzheimer’s disease have a steady decrease in memory and other cognitive abilities. Graph-based models are widely used to detect EEG-based Alzheimer's disease. In [48], GNNs have been empirically evaluated and shown to effectively classify Alzheimer’s disease by analyzing functional connectivity. A novel approach, introduced in this paper, involves a dynamic spatial graph convolutional neural network (ST-GCN) that leverages both spatial topology and dynamic temporal information through 1D convolution for AD diagnosis. Additionally, the Adaptive Gated Graph Convolutional Neural Network (AGGCN) described in [72] targets AD by enhancing node features through convolution and employing a correlation-based functional connectivity metric to adapt graph architectures effectively.

D. MOTOR IMAGERY TASK

Motor imagery brain-computer interfaces, primarily used in medical applications, benefit from advancements in GNN technologies to enhance patient monitoring with movement issues. In [101], a GNN approach is utilized to decode time-resolved EEG signals during various motor imagery (MI) activities, focusing on the functional topological connection of electrodes. Additionally, [99] introduces a novel method to simulate efficient connections between brain regions for motor imagery identification using EEG inputs and GNNs. Further developments in this field include the introduction of a Double-branch graph convolutional neural network (DGCAN) in [77], which filters channels minimally influenced by spatial variables and employs spatial-temporal domain convolution to refine and extract features, effectively reducing individual variations influenced by geographic factors. In [128] proposes a multi-scale residual network with hybrid attention (MSHANet) to decode four motor imagery classes.

TABLE V
BRIEF SUMMARY OF GRAPH NEURAL NETWORK APPLICATION ON EMOTION RECOGNITION

Reference	Publication	Model	Data	Classification Accuracy (Mean ± Std)
[90]	Journal	DGCNN	SEED	SEED: (Subject-dependent: 99.25%, Subject-independent: 98.51%)
			DEAP	DEAP: (Subject-dependent: 98.96%, Subject-independent: 98.31%)
[42]	Journal	MST-GNN	DEAP	Accuracy: 92.31%±5.34%
[49]	Journal	GNN	SEED	Accuracy: 98.35%
[44]	Journal	SOGNN	SEED	Average Accuracy: 86.81±5.79%
			SEED-IV	Average Accuracy: 75.27±8.19%
[45]	Journal	RGNN	SEED	Subject-dependent Accuracy: 94.24±05.95% 79.37±10.54%
			SEED-IV	Subject-independent Accuracy: 85.30±06.72%, 73.84±08.02%
[53]	Journal	ResGAT	DEAP	Arousal: 87.06%, Valence: 89.26%,
				SEED-IV: 97.73%
[55]	Journal	STGATE	SEED	Accuracy: 90.37%
			SEEDIV	Accuracy: 76.43%
				Accuracy: 76.35%

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			DREAMER	
[59]	Conference	GFT-STANN	DEAP	Valence: 94.8%, Arousal: 96.1%, Four Class: 92.7%
[61]	Journal	SS4-STANN	DEAP	Valence: 95.6%, Arousal: 97%
[80]	Journal	PGCN	SEED-IV SEED-V MPED	Accuracy: 76.96±07.9% Accuracy: 71.40±09.43% Accuracy: 28.39±05.02%
[74]	Journal	FGCN	SEED SEED-IV	Accuracy: 94.1% Accuracy: 77.14±15.71%
[78]	Journal	MRGCN	DEAP SEED	Arousal: 95.72%, Valence: 94.97%, SEED: 98.9%
[103]	Journal	MdGCNN-TL	SEED DEAP	SEED: 91:54% and 85:01% DEAP: 86:67% and 70.19%
[104]	Journal	DBGC-ATFFNet	SEED SEED IV DEAP	SEED: 97.31±1.47% SEED IV: 89.97±2.85% Valence: 95.91±3.27%, Arousal: 94.61±2.89%
[82]	Journal	GECNN	SEED SDEA DREAMER MPED	Average Accuracy: 92.93% Average Accuracy: 79.69% Average Accuracy: 95.73% Average Accuracy: 40.98%
[83]	Conference	GC-GCNN	DEAP	Arousal: 90.11%, Valence: 89.48%, Arousal- Valence: =86.35%
[92]	Journal	P-GCNN	DEAP, SEED	Valence: 73.31%, Arousal: 77.03%, Dominance: 79.2% SEED: 84.35%
[93]	Journal	CGCNN	SEED SEED-IV	Average accuracy: 93.36% Average accuracy: 75.48%
[98]	Journal	GCNN	DEAP SEED	Average Accuracy: 90.74%, Average Accuracy: 91%, Average Accuracy: 90.22%
[100]	Journal	STFCGAT	SEED DEAP	SEED: (Subject-dependent: 99.11%±0.83%, Subject-independent: 94.83%±3.41%) DEAP: (Arousal: 91.19% ± 1.24%, Valence: 92.03% ± 4.57%)
[107]	Journal	MDGCN-SRCNN	SEED SEED-IV	Average Accuracy: 95.08% Average accuracy: 85.52%
[108]	Journal	Mul-AT-RGCN	DEAP	Valence: 93.19%, Arousal: 91.82%
[110]	Journal	GCNN	DEAP	Valence: 90.45%, Arousal: 90.60%
[87]	Journal	DGCNN	SEED DREAMER	SEED: (Subject-dependent: 90.4%, Subject-independent: 79.95%) DREAMER:(Valence: 86.23%, Arousal: 84.54%, Dominance: 85%)
[64]	Conference	Dynamic GNN	SEED	Average Accuracy: 94.35±5.05%
[119]	Journal	DAMGCN	DEAP SEED SEED-IV	DEAP: (Valence: 96.96%, Arousal: 97.17%, Dominance: 97.50%), SEED: 99.42%, SEED-IV: 96.86%., 90.4±2.61%, 83.55±3.88%
[122]	Journal	SOGPCN	SEED	Subject-dependent = 95.26%, Subject-independent = 94.22%

Table V summarizes various papers on graph-based networks for EEG-based emotion recognition. It details research methods, newly developed GNN variants, and their applications. The table also provides publication specifics, such as article types and years, along with the achieved results. This overview offers valuable insights into the advancements and effectiveness of graph-based networks in EEG-based emotion recognition.

E. FATIGUE DRIVING RECOGNITION

Fatigue detection using EEG signals presents challenges in noisy environments, where traditional models fall short. Graph-based networks have gained prominence in EEG fatigue detection. In [58], a deep graphical neural network-based framework was introduced, capable of denoising EEG signals and dynamically constructing functional brain networks. Another innovative approach in [79] is the fatigue detection technique using temporal and graph convolution, termed MATCN-GT. This model integrates a Multi-Scale Attentional Temporal Convolutional Neural Community block (MATCN block) and a Graph Convolutional-Transformer block (GT block). The MATCN block processes EEG signals directly without prior knowledge, while the GT block analyzes functions between specific electrodes, employing a multiscale attention module to preserve critical information about electrode correlations.

F. SLEEP STAGE CLASSIFICATION

Sleep stage classification is critical in sleep evaluation and illness diagnosis. Long-term sleep difficulties can lead to a variety of disorders. The EEG changes to varied degrees throughout sleep and varies with sleep depth. Sleep is categorized into several categories based on different EEG features. Automatic sleep stage classification has significant therapeutic significance and may substantially increase the efficiency of traditional sleep stage classification. A wide number of scholars have made significant contributions to the automation of this categorization problem. Notable contributions include [96], where BayesEEGNet, a Bayesian

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graph neural network, models brain activity as a Poisson process with connection probability diagrams for autonomous sleep stage classification. In [113], GraphSleepNet, a deep-learning framework, uses adaptive learning to refine a spatial-temporal graph convolutional network (ST-GCN) for EEG-based sleep stage classification. Additionally, [116], introduces a novel approach using a Jumping Knowledge Spatial-Temporal Graph Convolutional Network (JK-STGCN) to classify sleep stages effectively.

G. NEUROLOGICAL DISORDERS DIAGNOSIS

Neurological disorders (ND) are nervous system illnesses that affect the brain, spinal cord, nerves, and muscles. Here, [129] different brain disorders create variations in the typical EEG image, which allows the nature of the disease to be identified. In [112, 126], a new graphical convolutional neural network (GCNN)-based technique to improve the diagnosis of neurological illnesses utilizing scalp electroencephalogram (EEG) is presented. In [102], a novel method for detecting neurological diseases from electroencephalogram (EEG) signals is presented, emphasizing the utility of graph neural networks in EEG signal analysis. This approach suggests that graph neural networks can effectively address a variety of EEG data processing and analysis challenges, offering potential for improved accuracy and efficiency in diagnosing neurological disorders.

H. OTHERS AREA

Graph-based neural network models are extensively across various aspects of EEG signal analysis, with numerous variants developed to meet specific demands, particularly in disease diagnosis. The attention-based sparse graph convolutional neural network (ASGCNN) is tailored for diagnosing Parkinson's disease, overcoming challenges associated with weak links and statistical limitations. In [73], C-GCN, an extension of GCN designed to extract temporal, frequency, and spatial features from EEG signals, is introduced, using coherence networks to classify motor imagery tasks in spinal cord injury. The segmental adaptive spatial-temporal graph convolutional network (SAST-GCN), detailed in optimizes P3-based video object identification by dynamically segmenting data according to P3 processing phases. An adaptive graphical neural network (AGNN) is used for classifying autism spectrum disorder (ASD) in children, focusing on abnormalities in the left inferior circuit. Here, [95] discusses a sequential learning approach using a graph isomorphic network (GIN) for interpreting EEG and EMG signals in stroke rehabilitation. Lastly, [130] explores the early use of GNNs in detecting EEG signal spikes to identify epileptogenic networks and enhance 3D source localization accuracy.

VI. DISCUSSION

This review paper is essential as it consolidates the rapid advancements in GNN applications for EEG, providing a structured overview that is critical for both new researchers and experts. It bridges the gap between diverse studies and synthesizes their findings to highlight effective strategies and pinpoint areas needing further exploration. Additionally, this paper will guide future research directions, fostering innovations that harness GNNs' full potential in clinical and research settings. GNNs are spearheading a revolution in EEG signal analysis, transforming our ability to detect nuanced neurological patterns and significantly enhance diagnostics in areas like emotion detection, seizure diagnosis, and stroke rehabilitation [131, 132]. GNNs offer a robust method for modeling intricate interactions between brain areas, marking a substantial improvement over traditional techniques and advancing our comprehension of neural networks in diverse cognitive states [133]. This paper also highlights the specific applications of GNNs in EEG signal classification and explores the various GNN variants tailored for EEG analysis. Extensive research utilizes GNNs for EEG-based emotion recognition, with specific variants and their impacts detailed across several tables and figures within the text. The paper includes comprehensive reviews of individual model architectures developed for different studies, showcasing their contributions to enhancing EEG analysis accuracy and efficiency, thereby opening new avenues for neuroscience and clinical research.

Additionally, it outlines the achievements of these variants in separate publications, as shown in Tables II, III, and IV. And Table V, Fig 3, illustrates the distribution of publications across various EEG-based applications.

A. LIMITATIONS OF GNNs

Graph Neural Networks (GNNs) have shown promising results in EEG-based research, yet several challenges need to be addressed to enhance their effectiveness and applicability. Generalizability remains a concern, as current models are often tailored to specific datasets or tasks, limiting their applicability across diverse EEG data and cognitive states [131]. To overcome this, future models should be tested across various datasets and tasks, with standardized benchmarks and evaluation metrics to assess their robustness. Interpretability is another critical issue, as the complex structures of GNNs make them "black boxes" [134]. Techniques such as attention mechanisms, feature importance analysis, and model simplification can improve transparency and trustworthiness, making them more acceptable in clinical settings.

Scalability is essential as EEG datasets grow in size, requiring GNNs to handle large graphs efficiently without compromising performance [135]. Approaches like graph sampling, hierarchical graph representations, and parallel processing can help scale GNNs effectively. Integrating EEG data with other neuroimaging modalities like fMRI or MEG poses challenges due to their heterogeneous nature [132]. Developing multi-modal GNN architectures that can fuse and exploit complementary information from different modalities can lead to more accurate and holistic brain analyses.

Temporal dynamics of EEG signals are crucial, yet current models need more sophisticated approaches to accurately capture these time-dependent features [136]. Incorporating techniques such as RNNs, LSTMs, and TCNs with GNNs can

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enhance the modeling of temporal dynamics. Personalization is important, as individual differences in brain structure and function are often not adequately addressed [137] Personalized GNN architectures that adapt to individual variations in EEG signals could improve accuracy and applicability.

EEG signals are prone to noise and artifacts, necessitating robust GNN architectures that can handle such imperfections [133]. Techniques like signal preprocessing, robust feature extraction, and noise modeling within the GNN framework can mitigate the impact of noise and artifacts. Transfer learning is limited in current GNN models, but developing strategies such as pretraining and fine-tuning can improve their versatility and reduce the need for extensive training data for each new task [138]. Hyper parameter tuning is crucial for optimizing GNN performance, but manual tuning is time-consuming and often suboptimal. Automated hyper parameter optimization techniques can help find optimal configurations more efficiently [139]. Real-time processing is essential for applications like BCIs, requiring GNN architectures that provide low-latency predictions [140]. Model simplification, quantization, and hardware acceleration can help reduce computational load and enable faster processing of EEG data.

Addressing these challenges will make GNN models more effective, interpretable, and applicable to a wider range of EEG-based research and applications, paving the way for advancements in understanding and leveraging brain activity.

B. CHALLENGES OF GNNs APPLICATION FOR EEG SIGNAL

GNNs face several challenges in EEG signal analysis and other applications. In epileptic seizure detection, GNNs often struggle with over-smoothing, which can obscure subtle yet critical changes in EEG patterns necessary for accurate detection. The variability in EEG signals among epilepsy patients further complicates the generalization of GNN models. In emotion recognition, GNNs must effectively process complex signal relationships across different brain regions. Constructing accurate graph structures that capture emotional states is difficult due to the subtlety of emotional expressions and individual differences. For Alzheimer's disease diagnosis, GNNs are required to identify EEG patterns indicative of the disease, which vary widely among individuals. The progressive nature of Alzheimer's also necessitates the dynamic capturing of EEG changes over time, challenging GNNs' ability to handle time-series data effectively. In fatigue driving detection, the presence of noise and artifacts in EEG signals can degrade GNN performance, while accurately capturing dynamic changes in fatigue states remains a significant challenge. To address these limitations, advancements in GNN models are essential, focusing on improving interpretability, generalization, and handling of EEG complexities. Enhanced data preprocessing and graph construction techniques are also crucial for boosting GNN performance in these areas.

C. LIMITATION OF THE REVIEWED PAPER

GNN model is a rapidly growing research area. They are still in the infancy of processing EEG signals, and there are still some limitations: (1) The majority of research conducted in the field of EEG analysis has primarily focused on offline experiments, with little consideration given to GNN applications in real-time EEG analysis. While the experimental out-comes are promising, there is a notable shortage of publications dedicated to addressing the unique challenges associated with EEG data analysis. Real-time applications introduce several complexities, such as the need for low-latency processing, robustness to noise and artifacts, and the ability to adapt to rapidly changing data streams (2) EEG signals are dynamic, both spatially and temporally, which poses significant challenges in their analysis. Many reviewed studies face limitations when dealing with complex EEG data, particularly as numerous GNN models struggle to capture the temporal dependencies that are essential for accurate analysis. Current GNN architectures typically rely on static graph structures, which are not well-suited to the time-varying nature of complex EEG data. Additionally, EEG data is often noisy, with artifacts that GNNs are not inherently equipped to handle, leading to potential reductions in accuracy, especially in real-time applications. Moreover, GNNs are computationally intensive, making it challenging to process large, high-dimensional EEG data in real time applications.

D. FUTURE RESEARCH

Graph neural networks have garnered significant attention in recent years due to the extensive research efforts dedicated to their development. In our review, we have primarily focused on examining the GNN model and its newly developed variants, along with their applications. We have provided comprehensive discussions on these variants and summarized their applications using tables and figures. Looking ahead, we intend to investigate deeper into the various components of GNNs, such as graph structure, node pooling techniques, convolution layers, node features, and graph embedding systems. Future work could concentrate on enhancing these models across several critical areas: (1) Enhancing model robustness and generalization: Develop GNN architectures that perform consistently well across diverse EEG datasets and various neurological conditions, adapting to differences in demographics, environments, and equipment. (2) Scalability to Larger Datasets: Address the need for GNNs to handle increasing EEG data volumes by developing scalable training methods and distributed computing strategies. (3) Improving Computational Efficiency: Optimizing GNNs to reduce their computational de-mands will make them more accessible and usable in real-time applications, such as in neurofeedback systems or during surgical procedures where rapid data analysis is crucial.

VII. CONCLUSION

This paper offers a comprehensive review of recent advancements in GNNs for EEG signal analysis, highlighting on their adaptations for EEG classification and applications. It concentrates on newly developed GNN variants tailored

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specifically for EEG data, reviews their applications across various types of EEG signals, and provides comprehensive details on new model variants, dataset specifics, publication trends, and analytical comparisons of related studies. The integration of GNNs into EEG research has significantly enhanced our understanding of brain function, improved diagnostic accuracy, and enabled the development of innovative applications. It provides in-depth insights into feature extraction, model variants, and prediction methodologies. We summarize key findings, applications, limitations and propose future research directions to guide emerging researchers in this field. By leveraging GNNs, there is potential to unlock the full capabilities of EEG-based analyses, facilitating advanced discoveries in brain research. Additionally, this paper identifies future research opportunities and addresses existing challenges within the field.

CONFLICT OF INTEREST
The authors declare no conflict of interest

AUTHOR CONTRIBUTIONS
All authors have their contributions based on their expertise. Most of the works have been done by S M Atoar Rahman under the supervision of Prof. Yu Guo and Prof. Hui Zhou. Prof. Hui Zhou suggested the topic name and idea. Prof. Ziyun Ding and Prof. Dingguo Zhang are involved in the revision of the paper. MD Ibrahim Khalil has helped with the paper collection, reference updates, revision and formatting. The authors are ordered by their aids to this paper.

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ABBREVIATION
GNN: Graph Neural Network
GGN: Graph-generative neural network
MST-GNN: Minimum spanning tree-based graph neural network
SOGNN: Self-Organized Graph Neural Network
RGANN: Residual Graph Attention Neural Network
STGATE: Spatial-Temporal Graph Attention Network with a Transformer Encoder
STANN: Spatial-Temporal Attention Neural Network
GCNN: Graph convolution neural network
GCN: Graph convolution network
AGGCN: Adaptive Gated Graph Convolutional Network
C-GCN: Coherence-Based Graph Convolutional Network
FGCN: Fusion Graph Convolutional Network
LGCN: Linear Graph Convolution Neural Network
MGGCN: Multi-Granularity Graph Convolution Network
DGCAN: Double-Branch Graph Convolutional Attention Neural Network
MRGCN: Multi-head Residual Graph Convolutional Network
STFCGAT: Spatial-Temporal Feature Fused Convolutional Graph Attention Network
MATCN-GT: Multi-scale Attentional Temporal Convolutional Neural Network
PGCN: Progressive Graph Convolution Network
ST-GCN: Spatial-Temporal Graph Convolutional Network
MdGCNN-TL: Multi-domain fusion Deep Graph Convolution Neural Network-Transfer Learning
DBGC-ATFFNet-AFTL: Dual-branch dynamic Graph Convolution Adaptive Transformer Feature Fusion Network with Adapter-Fine-tuned Transfer Learning
GECNN: Graph-Embedded Convolutional Neural Network
GC-GCN: Granger Causality-Graph Convolution Network
Mul-AT-RGCN: Multimodal Attention Recurrent Graph Convolutional Neural Network
SAST-GCN: Segmentation Adaptive Spatial-Temporal Graph Convolutional Network
SSGCNet: Sparse Spectra Graph Convolutional Network
CGCNN: Causal Graph Convolutional Neural Network
LGCN: Linear Graph Convolution Network
EGCN: Epilepsy EEG Graph Convolutional Network
SGCN: Sequential Graph Convolutional Network
GICN: Graph Input layer attention Convolutional Network
MASS: Montreal Archive of Sleep Studies
HGCN: Hierarchy Graph Convolution Network
DGCNN: Dynamical Graph Convolutional Neural Networks
TSVM: Transductive Support Vector Machine
BRC: Brain Research Center
CAGNN: Connectivity-aware graph neural network
JK-STGCN: Jumping knowledge spatial-temporal graph convolutional network. F-FGCN: Forward-forward graph convolution network
DAMGCN: Dual Attention Mechanism Graph Convolutional Neural Network
GERBN: Graph emotion recognition with brain functional connectivity network
SOGPCN: self-organized graph pesudo-3D convolution

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